

**Period**

A set of elements in a row in the periodic table. Each atom of the elements in the first period have one electron shell. Atoms in the second period have two electron shells.

**Periodic table**

A table that identifies and classifies all known elements.

**Photograph negative**

A film or plate that has been exposed to light to show an image with reversed colours.

**Photosynthesis**

A complicated set of chemical reactions by which plants make their food. They use the energy of sunlight to convert water and carbon dioxide into sugar and oxygen.

**Pollution**

Harmful substances that are released into the environment. Pollution can be chemical – a gas, liquid, or solid added to the air, water, or soil.

**Pressure**

A measure of how hard a force pushes on a surface. Pressure depends upon the strength of the force and the area of the surface.

**Proton**

A positively charged particle in the nucleus of an atom. Protons attract electrons, and they circle the nucleus.

**Radar**

A system that detects the positions and speed of distant objects, such as aircraft.

**Radiation**

Energy released by atoms as light, infrared, ultraviolet (UV), and X-rays. The term “radiation” is also used to describe the rays given out by radioactive substances.

**Radioactive**

Describes a substance that contains unstable atoms, in which the nucleus breaks apart,

or decays. When a nucleus decays, it releases at least one particle and its atomic number changes, so that the atom becomes a different element.

**Reaction**

A chemical process in which atoms or molecules form bonds with each other to form a new compound – or rearrange, to produce different compounds.

**Renewable energy**

A source of energy that will not run out, for example, wind.

**Rust**

The common name for the compounds that form when iron reacts with oxygen and water.

**Salt**

A compound that forms when an acid reacts with an alkali. Sodium chloride is the most familiar example of a salt.

**Semi-metal**

An element that has properties of both metals and non-metals.

**Smelting**

A chemical process that involves great heat in which a metal is extracted from its ore.

**Solid**

A state in which the particles of matter (atoms or molecules) are bound to each other, so they remain in fixed positions. A solid has a fixed shape and volume.

**Stainless steel**

An alloy of iron and carbon that also contains other metals, such as chromium, which stop the iron from rusting and increase its toughness.

**Superconductor**

A substance through which an electric current flows with no resistance; most substances resist the flow of electricity, and produce heat when current flows.

**Synthetic**

Another word for artificial. Out of the 118 known elements on Earth, more than 25 are synthetically made.

**Toughness**

A quality of a solid that shows how difficult it is to break. Steel is very tough – it may bend or twist, but is much harder to break.

**Transition metal**

A member of the set of metals that sit in the middle of the periodic table. Most metals belong to this set.

**Toxic**

Another word for poisonous or harmful.

**Transparent**

Another word for see-through. Glass, water, and air are transparent to light. Many materials are transparent to other forms of radiation.

**Vacuum**

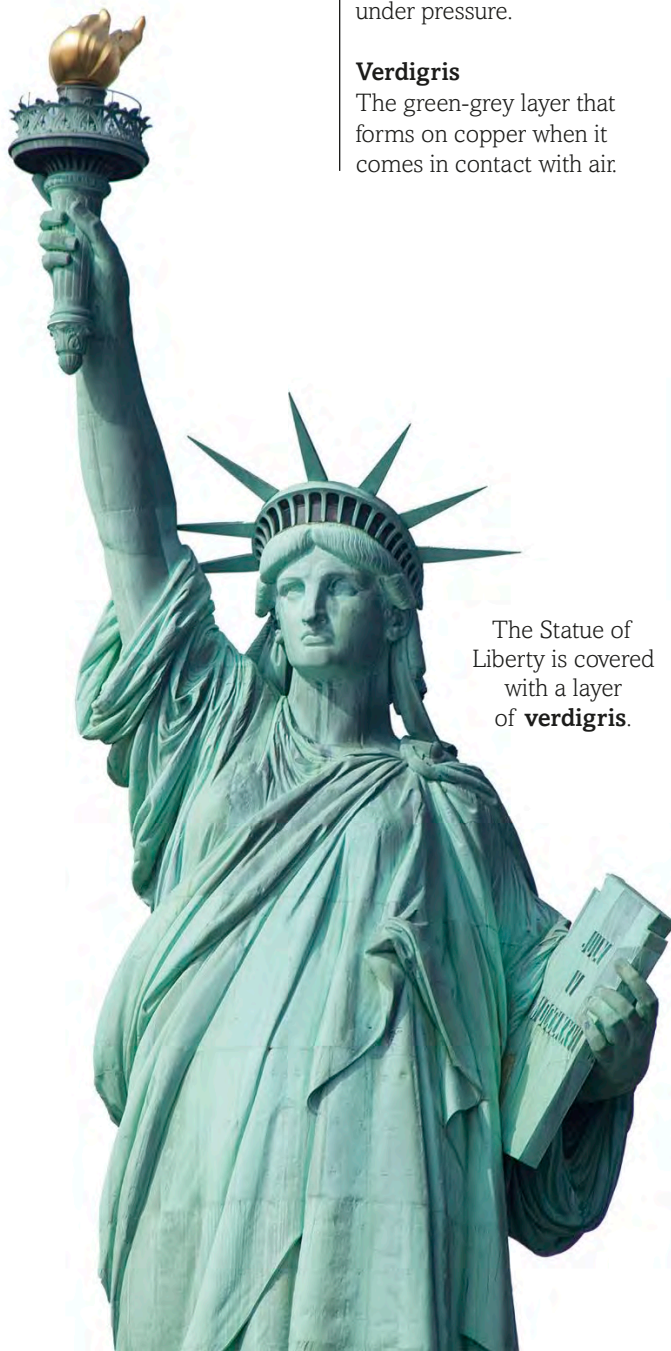
An empty space that does not contain air or any other substance.

**Vapour**

A gas that can easily be changed back to a liquid, by cooling it or putting it under pressure.

**Verdigris**

The green-grey layer that forms on copper when it comes in contact with air.



The Statue of Liberty is covered with a layer of **verdigris**.